

Title Blockchain –

Task 2. Smart Contracts as Legal Contracts

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Table of contents

Contents

Introduction	1
Challenges of considering smart contracts as legal contracts.....	3
Jurisdiction.....	3
Legality.....	3
Scalability.....	5
Security	5
Immutability	6
Data privacy laws	6
Lack of Standards	7
Complexity.....	7
Enforceability	7
Relying on external data	8
Approaches that may be used to solve challenges.	8
Private Blockchain.....	8
Cryptography	9
Global standards	9
Security	9
Types of application scenarios where smart contracts can adequately be interpreted as legal contracts.	9
Retail payments.....	10
Real Estate	10
Insurance	11
Agribusiness	11
Conclusion	11
Bibliography	13

List of abbreviations (if necessary)

GPDR – General Data Protection Regulation

HIPA – Health Insurance Portability and Accountability Act

CCPA – California Consumer Protection Act of 2018

ISO – International Standards Organization

ITU– International Telecommunication Union

Introduction

Traditional Contracts have been around since as early as ancient civilization. They have been used to enforce rights and obligations between two or more parties. Typically contracts involve the transfer of goods, services or money or promise to transfer any of them in the future("Contract," 2024). This contract can be as simple as verbal or complex where there needs to be third parties with deeper understanding in law engaged to draft them. As a result, some of the contracts can be very expensive to execute due to the complexity of the transaction involved. Due to jurisdiction issues its very expensive for individuals and small organization to enforce this traditional contract across multiple jurisdictions. Anchored on the legal system and laws drafted and adopted by various jurisdictions, the traditional contracts are enforceable in courts in various jurisdictions. Most of these contracts are drafted and executed by the parties involved, most of which are paper-based or electronic contracts. This is susceptible to tampering. (*Smart Contracts Vs Traditional Contracts*, 2023).

To be able to execute the traditional contract, most jurisdictions require that there is proper identification of the parties. The identification comes in the form of government documents that prove identity. Without identification the contract can face legal challenges when it comes to enforcement and registration as a binding contract. According to the world bank just about 1 billion people in the world don't have identification(*1.1 Billion 'Invisible' People without ID Are Priority for New High Level Advisory Council on Identification for Development*, n.d.). This makes them ineligible to enter legal contracts. This might not change very much as more people in third world countries find it more difficult to access this documents due to inefficient and corrupt systems.(mwendwa, 2020)

Despite Traditional contracts being around for a long time. They fall short when it comes to execution at time. This has resulted in multiple lawsuits and becomes very costly to enterprises to enforce despite long hours and time used to prepare this contracts.("Infosys Loses Multi-Billion Deal," 2023)

As we discuss the traditional contracts and the limitations that it has over the years, there has been development of different technologies that have been coming together to revolutionize how contracts are executed. The first is the spread, penetration, and accessibility of fast internet. As of January 2024 about 66.2% of the worlds population has access to the internet(*Global Internet User Penetration 2024*, n.d.), This enables devices (Oracles) that collect information and feed to blockchain technologies to be deployed more remotely and widely.

Second is the developments around blockchain technology, Blockchain is a shared immutable ledger that facilitates the process of recording transactions and tracking assets in a business network(*What Is Blockchain Technology?*, n.d.). Blockchain can also be described as an immutable ledger of transactions shared among mutually untrusted parties(Falazi et al., 2019, p. 3). There are other factors that have also contributed to enable access like the advancement and lowering of costs in the Solar energy sector which has enabled access to electricity to power devices such as IOT devices or Oracles.

The above technologies have enabled the development of smart contracts. (Szabo, 2017, p. 5) provides two definitions for smart contracts. The first one is as a machine with rules that could have been defined in a contract but instead wrote into a machine, he also describes smart contracts as an application that runs in a distributed and trust-minimized manner on a blockchain.

(Szabo, 2017, p. 8) defines three major benefits for smart contracts, reducing mental transaction costs, increasing predictability and security. Unlike the traditional contract smart contracts are automated. This makes them execute automatically depending on the programmed tasks. Smart contracts deployed on blockchain technology which enables them to be decentralized and trustless. Due to the tokenization capabilities on blockchain smart contracts can be used to trade by integrating with digital assets such as crypto tokens.(*Smart Contracts Vs Traditional Contracts*, 2023)

Smart contracts differ from traditional contracts.

Transparent – All parties who are part of the blockchain can always view the contract terms. traditional contracts on the other hand have restrictions on access and sometime there is lengthy bureaucracy to get access to the contracts.

Immutability – Because they are deployed on blockchain they cannot be altered or deleted once they have executed. Traditional contracts on the other hand can be subject to alteration by replacement. Forgery or alteration

Automated – This are computer programs that self-execute once certain conditions are met. Traditional contracts require a different mechanism to enforce and sometimes interpretations can lead to disputes.

(*Smart Contracts Vs Traditional Contracts*, 2023)

In this paper we consider the challenges of considering smart contracts as legal contracts and the approaches we can use to resolve the challenges. After which we will look at some of the application scenarios where smart contracts can be interpreted as legal contracts.

Challenges of considering smart contracts as legal contracts

Jurisdiction

Smart contracts are based on blockchain technology. Decentralization with no single point of control for this technology might make it complicated to adjudicate disputes that might arise from smart contracts. Global adoption means that the parties concerned may be executing the contract across borders which might not be accommodated in the laws one every jurisdiction. (Lipton & Levi, 2018). Traditional contracts specify the jurisdiction and governing law where someone might seek legal reprieve in case there is need. Smart contracts are executed on blockchain. There could be issues raised on whether the node that executed the smart contract can legally be adjudicated in certain areas especially where sanctions have been applied to some countries.

Legality

Are smart contracts legal? Governments around the world have started to come up with laws to regulate smart contracts. This is a process still undergoing evolution. The nature of smart contracts is still evolving. (Szabo, 2017, p. 5) Compares a contract to a vending machine, which would produce dispense a soda when money is inserted into the machine. In this case the contract is encoded in the machine's gears. This is easy to interpret as compared to smart contracts which are code. To be legal, smart contracts should adhere to the laws and regulations in the jurisdiction where they can be enforced. To facilitate this, they must be anchored in the laws of jurisdiction. (Allen & Hunn, 2022, p. 34) outlines two important point on when smart contracts could be seen as legal, i) Current hailed "smart contracts" are mainly performance mechanisms hence would not stand as legal contracts. ii) Smart contracts cannot exist in a legal vacuum. some jurisdictions to define ant entrench smart contracts into law. The state of Arizona goes ahead to define a smart contract as "event-driven program, with state, that runs on a distributed, decentralized, shared and replicated ledger and that can take custody over and instruct transfer of assets on that ledger"(44-7061 - *Signatures and Records Secured through Blockchain Technology; Smart Contracts; Ownership of Information; Definitions*, n.d.). Such legislation can help to anchor smart contracts in law and make them legal. While this is a

beginning many countries are yet to define and recognize smart contracts as agreements that can either work alongside or replace traditional contracts.

There are cases where dispute resolution of smart contracts has end up in the legal system for adjudication (Novosibirsk State University & Zainutdinova, 2021, p. 1630) refers to B2C2 Ltd v. Quoine Pte Ltd in which case smart contract was contract of exchange of crypto currency where a Singapore court ruled in favor of the plaintiff. This means that despite the transactions conducted by smart contracts, Legal courts in jurisdictions can still nullify them.

We will now address some of the legislations that have attempted to recognize smart contracts as legal contracts.

Arizona HB2417 Explicitly permits smart contracts. “Smart contracts may exist in commerce. a contract relating to transaction may not be denied legal effect, validity, or enforceability solely because that contract contains a smart contract term”. This removes any doubts that smart contracts are not legal contracts.

In Vermont, Vermont H.868 12 V.S.A 1913 recognizes that “A digital record electronically registered in a blockchain, if accompanied by a declaration that meets the requirements of subdivision (1) of this subsection, shall be considered a record of regularly conducted business activity pursuant to Vermont Rule of Evidence 803(6) unless the source of information or the method or circumstance of preparation indicate lack of trustworthiness.” In this category, smart contracts data can be interpreted as part of legal contract and can be used in a court of law

Smart contracts can act as can be used alongside traditional contracts as tools of enforcement. An example is where internet service providers provide services level agreements to the customers. In such cases there is a thresh hold where the customer gets service credit. Below is an example.

Service Availability in any calendar month	Service Credit
$\geq 99.99\%$	No Service Credit
$\geq 99.85\%$ and $< 99.99\%$	1 Service Credit
$\geq 99.50\%$ and $< 99.85\%$	2 Service Credits
$\geq 99.00\%$ and $< 99.50\%$	3 Service Credits
$\geq 97.85\%$ and $< 99.00\%$	6 Service Credits

$\geq 90.0\%$ and $< 97.85\%$	10 Service Credits
$< 90.00\%$	15 Service Credits

This can easily be implemented as smart contract alongside the traditional contract on blockchain.

Scalability

Smart contracts are executed on blockchain chain. There are scalability issues associated with blockchain technology that affects smart contracts execution and completion. Some blockchain technologies such as Ethereum and Bitcoin adopt proof of work as a method to create a distributed trustless consensus to mine a new block. Miners compete to solve a difficult mathematical puzzle. A block can only be submitted to a block chain with the correct answer. This proof of work is resourceful and time consuming however its easy for others to verify. To address specifically the Ethereum network, one of the biggest scalability is that every node has to process all the transactions and has to store the entire state of the blockchain.(Scherer, n.d., p. 21). The other challenge is the reward mechanism used in blockchain technology. In Ethereum gas is used and as the transactions grow the costs go high. This could limit the ability of users to execute the contracts on the blockchain.

One of the methods that has been used to improve scalability is sharding. Instead of all the nodes processing all the transactions. A few selected nodes can be used to execute and verify transactions. The other option is the optimization of code. Another solution suggested is having both on chain and off chain execution of smart contracts. The mixing could optimize what we have to put on the blockchain while not compromising the transaction.

Security

Smart contracts deployed on various blockchain have exhibited security vulnerabilities. The more value that is held by smart contracts the more it becomes a target for hackers. Some of this vulnerabilities can lead to attackers manipulating transactions to launch intrusion attacks by publishing malicious contracts which can lead to collecting more digital assets on a single transaction (Samreen & Alalfi, 2021, p. 2)

In 2016 a DAO hack caused loss of 3.6 million ETH, This resulted in the infamous hard fork on Ethereum to recover lost assets (Thummavet, 2023)

Different blockchain platforms have different vulnerabilities. Before choosing a platform and deploying smart contracts its vital that we analyze the vulnerabilities specific to each platform to avoid deployment of code that can be exploited(Hu et al., 2019, p. 12)

Immutability

Blockchain technology does not allow for modification or deletion of previous transactions on the blockchain. Altering one block would break the whole chain. In the case where smart contracts have been deployed to the blockchain modification is not possible. This will make it difficult to correct errors or make amendments on the contract if needed(*Novosibirsk State University and Zainutdinova - 2021 - Legal Issues of Smart Contracts in Contract Law.Pdf*, n.d., p. 1630)

Data privacy laws such as GDPR give the person the right to rectify data and the right to be forgotten. The immutable nature of blockchain technology poses a challenge for this since its difficult or impossible to delete data/contracts stored on the blockchain.

In GDPR the responsibility to ensure the data is secure and abides by the regulations is the responsibility of the data controller and data processor. With no central authority when it comes to blockchain it could be difficult to comply with these laws.

Some have suggested that we should include a self-destruction code for smart contracts. This could be a solution but still fails to adhere to some privacy laws such as amending records.

To solve the problem of immutability and not being able to modify smart contracts. There needs to be a method to return transactions under conditions verified in a smart contract. This can be provided in the form of mutual restitution.(Novosibirsk State University & Zainutdinova, 2021, p. 1630)

Data privacy laws

Misuse of data has led to adoption of data privacy laws in various countries. A good example is the GDPR adopted by the European union. The decentralized nature of blockchain where smart contracts are deployed present a further challenge and makes it difficult do adhere to some of the data privacy laws. In the United States HIPA create standards on how healthcare providers can handle personal data, data manipulated on blockchain by smart contracts risk exposure given the transparent nature of blockchain(Kakavand et al., 2017, p. 8). CCPA gives consumers power of data that is collected about them and includes the right to delete the data. These laws would create a challenge given the architecture of smart contracts deployed on blockchain which does not allow for deletion of previous stored data.

In (*United States v. Gratkowski*, 2023) The defendants were charged with distributing child pornography using the bitcoin chain. This raised issues around the anonymity of blockchain and if it would allow illegal transactions. To balance between anonymity of the parties could bring issues of legality and data privacy to the center of legality of smart contracts on public blockchain.

Lack of Standards

Unlike other internet related technologies there are no uniform standards that would define movement of data and execution of smart contracts across jurisdiction. Different blockchain implementation make it difficult to interoperate between the platforms. Lack of common standards across jurisdictions has led to adoption of on non-standard laws that would make it difficult for to apply smart contracts across jurisdictions.

Complexity

Smart contracts are normally written as code and deployed on blockchain. The more complex the tasks that it needs to perform the longer and more complex the code will be. This would make it very difficult for a person who does not have skills in the programming language to read and interpret the code. At times this code can only be read by computer if deployed in compiled form. (Madir, 2018, p. 10)

The conversion of traditional contracts to smart contracts needs to be done carefully. The essence of blockchain technology is centered alongside elimination of 3rd parties by decentralizing authority. However, for one to translate the terms of the contracts to smart contracts would need knowledge of programming language. In the future the implementation of smart contracts needs to be simplified and allow for easy understanding and deployment. The current state it seems that instead of removing the central authority, it does introduce an additional authority that is the programmer and the lawyer.

Enforceability

It would be easy if the interpretation of law was based on clear and hard corded in black and white. In most cases the courts tend to rely on evidence that is beyond reasonable doubt. Beyond reasonable doubt might not be the same for every person. Sometime judgements have been overturned because the lower courts erred in considering the evidence. Smart contracts tend to just be a set of rules and actions that are coded in a programming language. It would be difficult for the this set of rules and actions would capture some of the probabilities in law. For example, what the law would interpret as a good faith or probable cause.

In the UK the law commission issued an advisory to the British government that smart contracts can easily be covered in the current legal framework. The advisory goes ahead and points out the already existing practices in online shopping where the goods are delivered, and payment is done. However, they point out that in the case where smart legal contracts are primarily deployed on code and deployed on blockchain, this could result in legal disputes (*Rise in Smart Legal Contracts Disputes Expected Following Law Commission's Review*, n.d.)

Relying on external data

In some cases, there is need to collect external data to execute as smart contract. This is through an external data source also known as the Oracle. In the case where there is a malfunction of the external device, and it provides inaccurate information then smart contracts might give unexpected results can have far reaching consequences. A good example would be the flight delay compensation. In the case where data is inaccurate might result massive compensation which might not be easily reversible

Approaches that may be used to solve challenges.

After we have looked at the challenges that smart contracts face on blockchain we now focus on some of the possible solutions that we can implement.

Private Blockchain

Private Blockchain can be used to address several challenges of smart contracts as legal contracts. The issue of scalability can be addressed by implementing a less resource intensive consensus algorithm that would validate the transactions faster and with less resources. For example, the Hyperledger reduces the cost of consensus by implementing the Practical Byzantine Fault-tolerant (PBFT) protocol. In some cases, the validators can be chosen by trust mechanisms which undermine the decentralization principle of blockchain technology.

In the case where data privacy laws require that users have more control over data, Article 17(1) of GDPR clearly provides that a data subject has the “right to be forgotten” by demanding the erasure of his/her personal data upon the withdrawal of consent, or upon his/her objections to the processing. Private blockchains can be used to implement this via forks or be programmed to delete user data after a specific period. Since the block is privately controlled there could be ways to amend the chain occasionally to accommodate data privacy requirements.

User data could also be stored off-chain and only hash values of the data stored on the chain. This would affect the performance of smart contracts and the transparency of blockchain technology.

There are technologies such as Zero-Knowledge prove that would allow sharing of data on the blockchain without exposing the data. This would ensure validation of the data without sharing the actual data. This can enhance security of the data.

Cryptography

One of the solutions is to encrypt data. Data stored on blockchain is encrypted. However, encryption makes it harder and not impossible to access the data. The other solution would be to use different techniques to mask the information of the persons involved in the smart contract, however this can create another problem that would make smart contract not eligible to work as legal contracts. (Kakavand et al., 2017, p. 9)

Global standards

Various international organizations such as ISO and ITU are coming up with international standards that will be able to provide for common implementation of blockchain technologies. This will be necessary in the success of smart contracts.

Security

Some of the approaches used to solve the security vulnerabilities of smart contracts include independent review of the smart contracts code. There are tools also being developed to analyze code and identify vulnerabilities. Adoption of best practices can help to prevent obvious vulnerabilities in smart contracts code.

Types of application scenarios where smart contracts can adequately be interpreted as legal contracts.

Some of the characteristics of blockchain have made the technology attractive for implementation in various industries. The cryptography, immutability of blockchain and decentralization which eliminates single point of failure. Participants in a blockchain share data hence making it transparent. The consensus mechanism that is used makes it difficult for hackers to manipulate the data.

In our introduction we have the issue of identification a how disenfranchising this is to people who cannot authenticate their identity. This is important for any transactions on a global scale. Blockchain provides a solution for one to be identified in an irreputable manner via cryptography. In addition to this there are now a lot of devices such as sensors, that conduct transactions on behalf of humans. Blockchain provides mechanisms to authenticate and identify this device. This can allow them to transact using smart contracts without centralized intervention.

We will now look at a few applications where smart contracts can be considered be adequately interpreted as legal contracts.

Retail payments

Since the Covid 19 pandemic there has been a lot of changes in the retail world. In developing countries this has led to many online retail apps that enable shoppers to shop for anything online. Uber has also its own retail application designated as Uber Eats.

This application involves simple transaction where users go online and shop for items in a store. Once they user has selected all the items and submits the order, a rider is selected to perform the delivery of the items. The rider picks the items from the store and delivers to the customer address provided. Once the customer receives the goods, they make the payments to the store which immediately pays the rider the transportation fee. This is an example where we can employ smart contracts to make settlements of the transactions.

Real Estate

One of the industries with a lot of intermediaries is the real estate industry. Most transactions involve real estate brokers, banks, lawyers etc. This makes the cost of real estate very expensive before the real payments for mortgages or rents are made.

Smart contracts can be used for identity management. Since data stored on the blockchain is completely secure and distributed it makes it hard for it to be stolen and tampered with

Smart contracts can be used to eliminate intermediaries by defining the terms and conditions on a smart contract. An example where a user wants to rent a house for a period of 3 months. This could be written in with a smart contract where once payment is made a key is generated over a 3-month validity period and sent to the customer. This would avoid a lot of paperwork involving the lawyer. Apart from this this

will also save costs and increase efficiency. Both customer and real estate owner can sign this contract digitally making it legally binding.

Insurance

One of the areas where smart contracts can really make a difference is the insurance companies. An example is building and construction sites where insurance companies can install IOT devices to monitor safety standards at the site. This would then be automating payments in case an accident claim is lodged against the insurance company.

To handle issues of privacy there is data that users would not want to provide to the insurance companies. However smart contracts can be used to hold this data and evaluate the data as per insurance policies. This would enable the insurance companies to settle claims without necessarily violating data privacy laws.

Agribusiness

Farmers enter into agreements with manufactures and distributors globally. It's difficult for the manufactures and distributors to guarantee quality and origin of supplies from farmers globally. There are international and local bodies that certify this product, however given the complexity and scale of operations, some end up subcontracting these services to less equipped firms that do not guarantee the global standards. There is lack of transparency in this process as there is no visibility. This leads to many problems including delayed payments to farmers. Smart contracts can be used to automate the whole process of tracking the goods from the farm to payments for the produce.

Conclusion

Smart contracts present opportunities for new innovations and disruptions in various industries such as logistics, real estate, Agribusiness. However, we have discussed limitations that smart contracts face from jurisdictions where different jurisdictions have different ways to interpret the law, scalability where discussed the limitations the underlying blockchain technology inhibit implementation of smart contracts, upcoming data privacy laws and how immutably aspect of blockchain technology will be affected by these new laws. We also discussed the enforceability of blockchain where courts of law have ambiguous decision-making criteria such as reasonable doubt. Other areas of concern include security and complexity aspects of smart contracts.

We also discussed some of solutions that might help in dealing with these challenges like the private blockchain which would help us navigate some of the data privacy laws, use of off-chain data, Global standards in the implementation of blockchain technology and finally security where we discussed independent reviewing of code by third parties and use of tools to detect vulnerabilities in code.

Finally, we discussed some applications where smart contracts can be interpreted as legal contracts. There is great opportunity and future research will need to address these issues to accelerate and make it easier accept and implement smart contracts.

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